Name
Date
Course

Rational Functions and Their Graphs (Part II) Homework

Graph the following functions. Draw asymptotes as dotted lines and write their equations.

① $f(x) = \dfrac{x+2}{-x+4}$

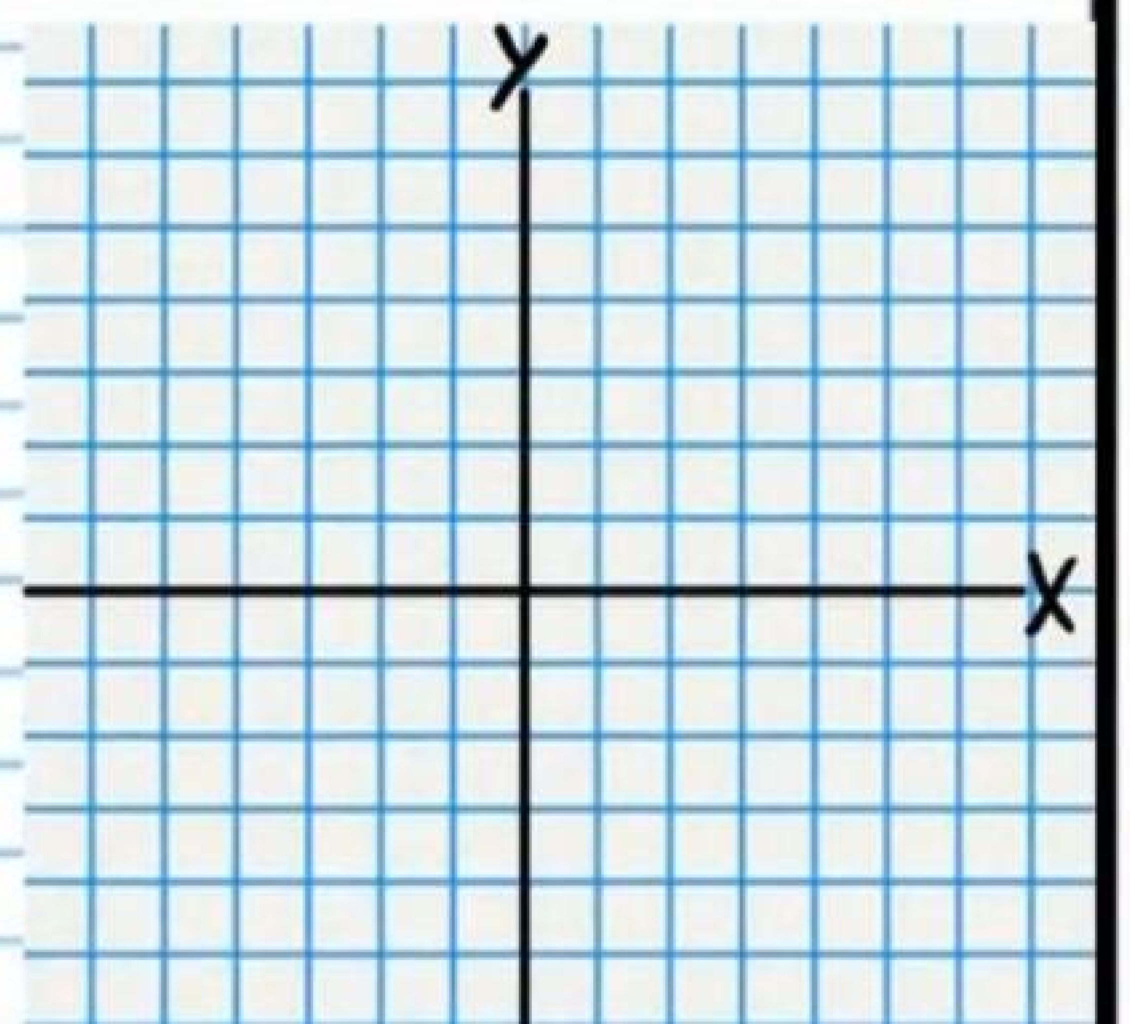
Horizontal Asymptote Equations

Vertical Asymptote Equations

Sign Chart

Y-Intercept

X-Intercept



② $f(x) = \dfrac{3x^2-10x-8}{x^2-4}$

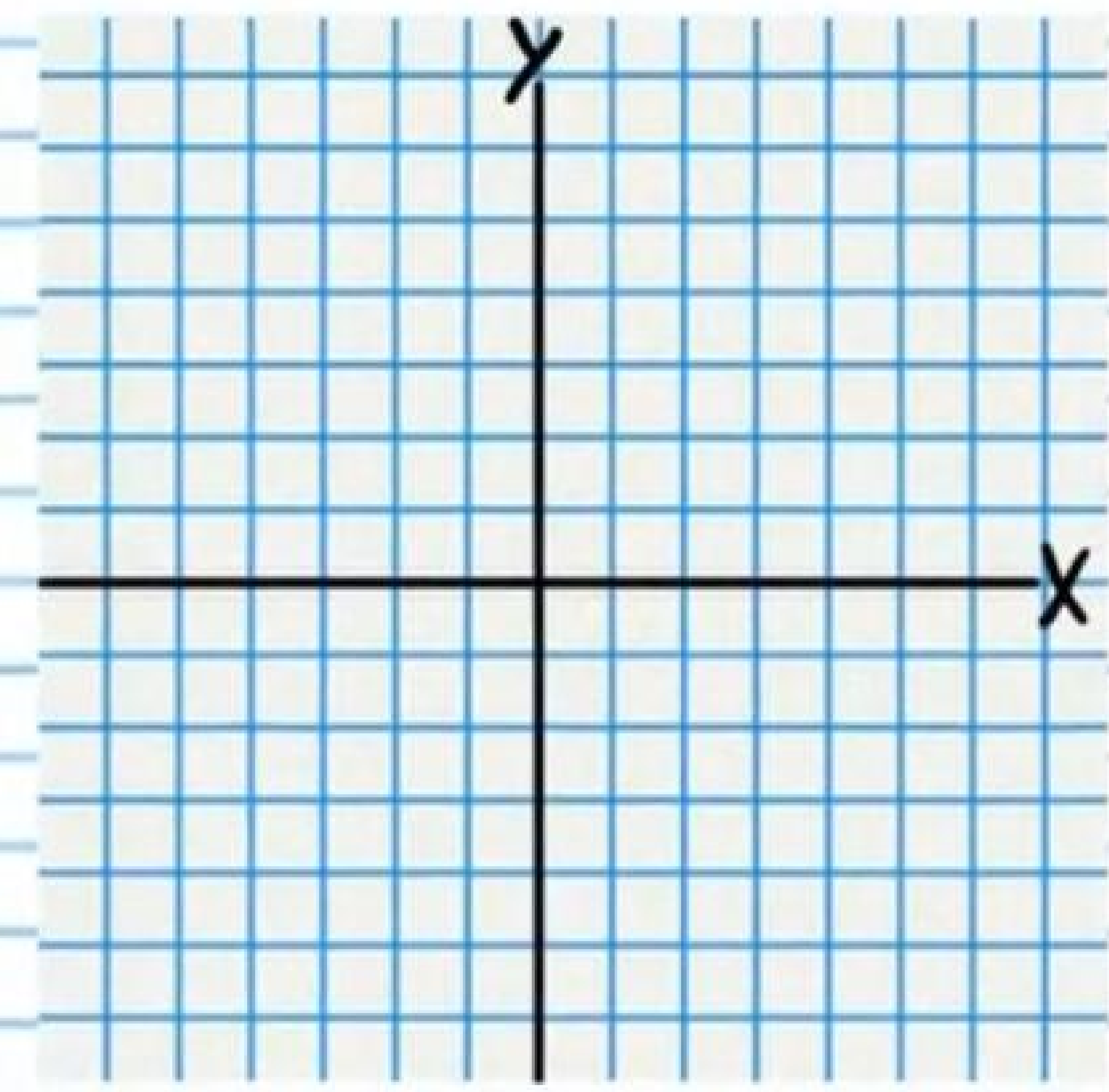
Horizontal Asymptote Equations

Vertical Asymptote Equations

Sign Chart

Y-Intercept

X-Intercept



③ $f(x)=\dfrac{x^4-1}{x^3-5x^2-x+5}$

Oblique Asymptote Equation

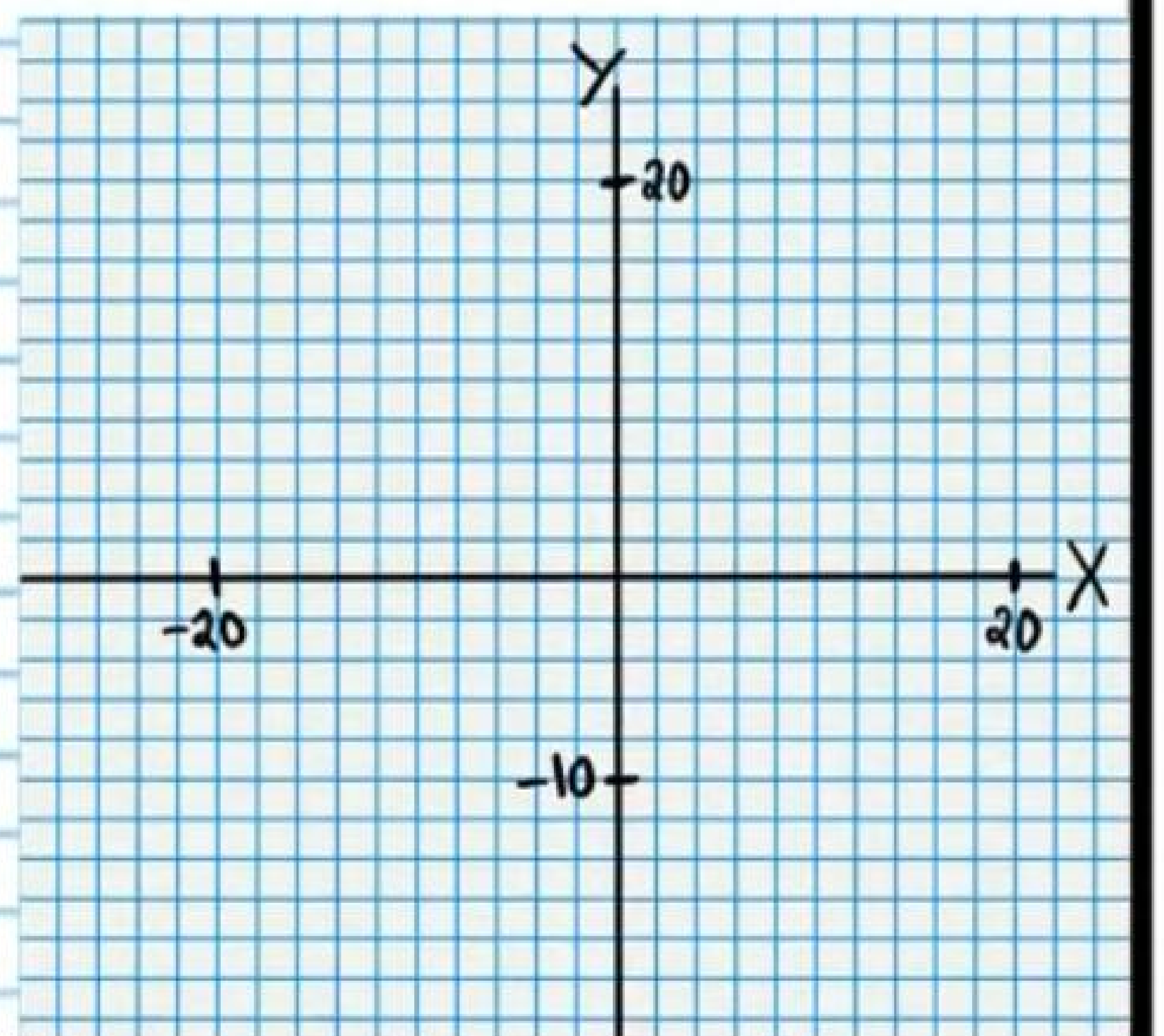
Hole(s)

Vertical Asymptote Equations

Sign Chart

Y-Intercept

X-Intercept



④ $f(x)=\dfrac{-x^2+3x-2}{x+3}$

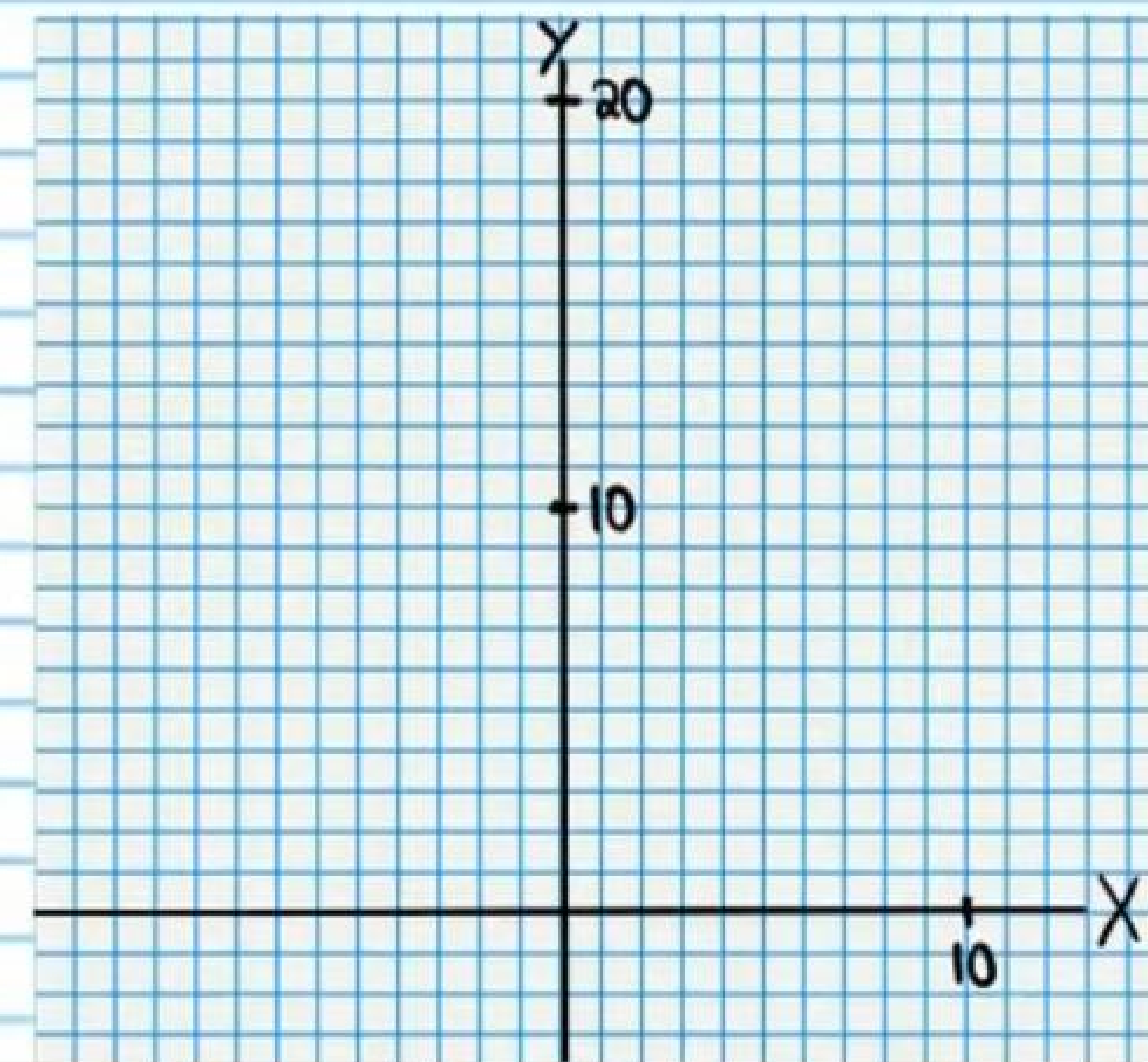
Vertical Asymptote Equations

Oblique Asymptote Equation

Sign Chart

Y-Intercept

X-Intercept



For each Function, Find the information requested.
It is not necessary to graph each function.

⑤ $f(x) = \dfrac{x^6+2x^5}{x^4+5x^3+6x^2}$

Hole(s)

Vertical Asymptote Equation(s)

Asymptote Curve

Y-Intercept

X-Intercept

⑥ $f(x) = \dfrac{x^3+3x^2+x+3}{x-3}$

Y-Intercept

X-Intercept

Vertical Asymptote Equations

Asymptote Curve

Answers

① $f(x) = \dfrac{x+2}{-x+4}$

Horizontal Asymptote Equations

$\boxed{y = -1}$

Vertical Asymptote Equations

$\boxed{x = 4}$

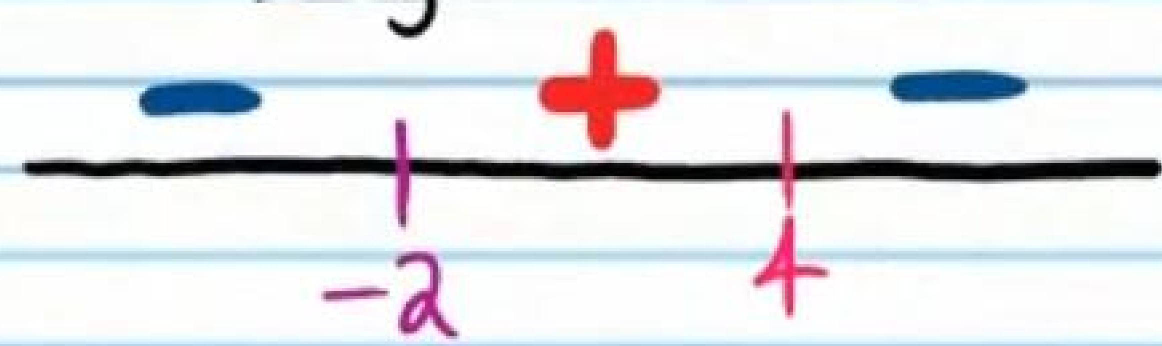
Y-Intercept

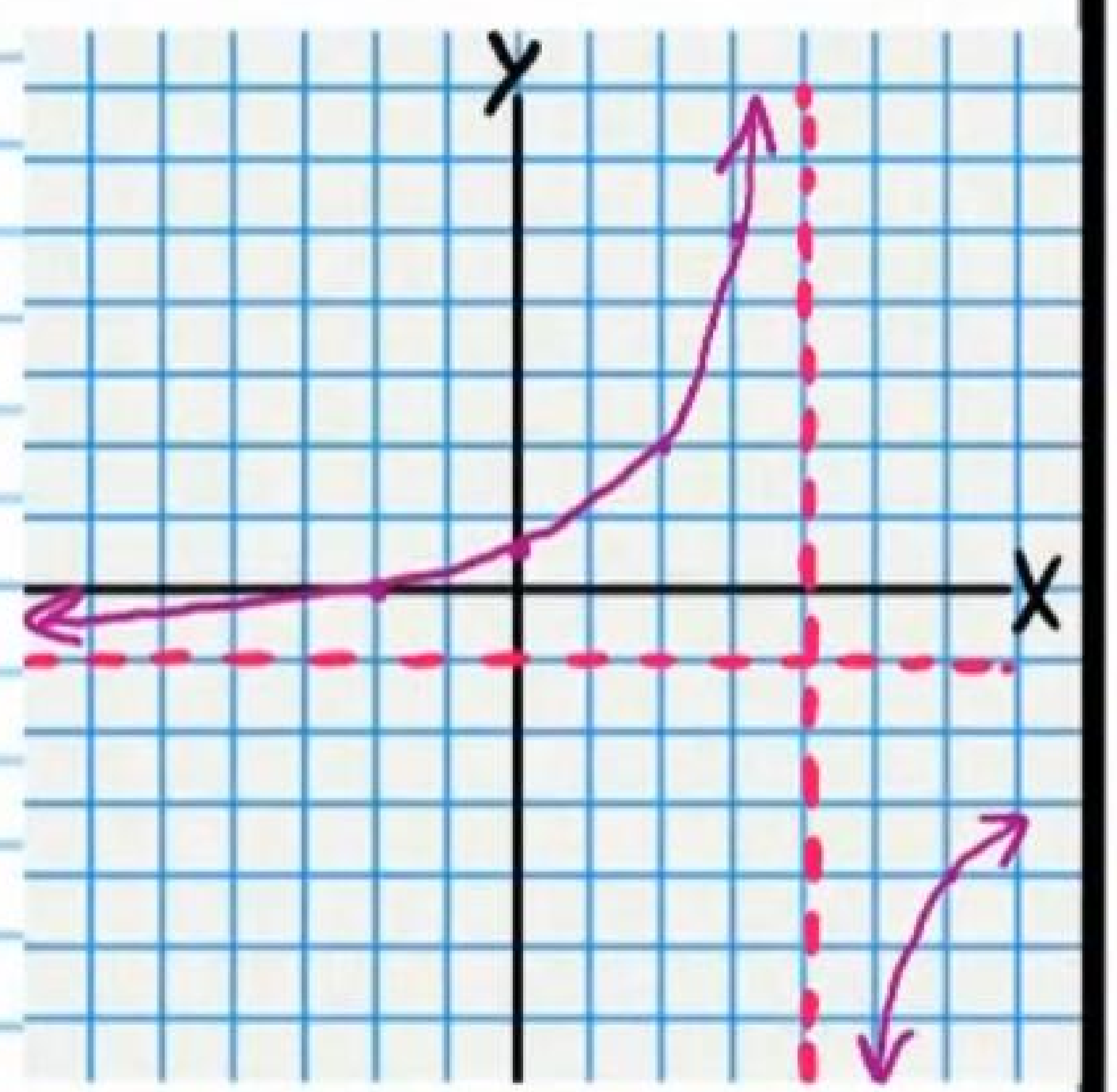
$\boxed{(0, \tfrac{1}{2})}$

X-Intercept

$\boxed{(-2, 0)}$

Sign Chart





② $f(x) = \dfrac{3x^2 - 10x - 8}{x^2 - 4}$

Horizontal Asymptote Equations

$\boxed{y = 3}$

Vertical Asymptote Equations

$\boxed{x = 2,\ x = -2}$

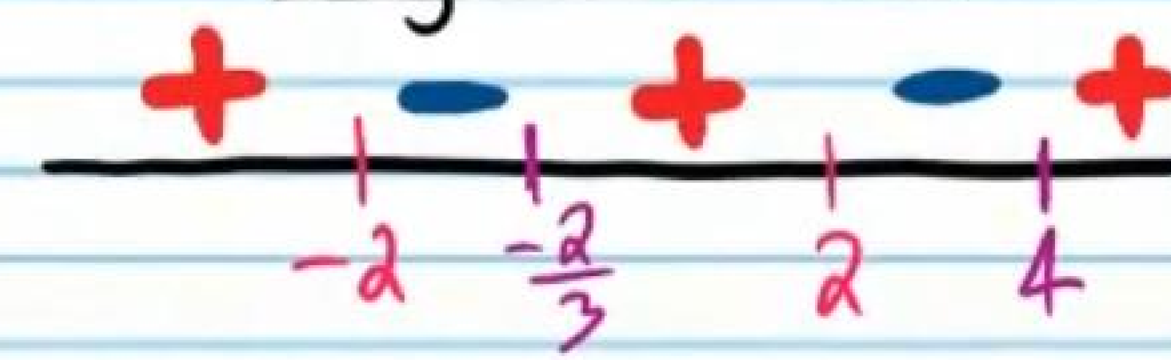
Y-Intercept

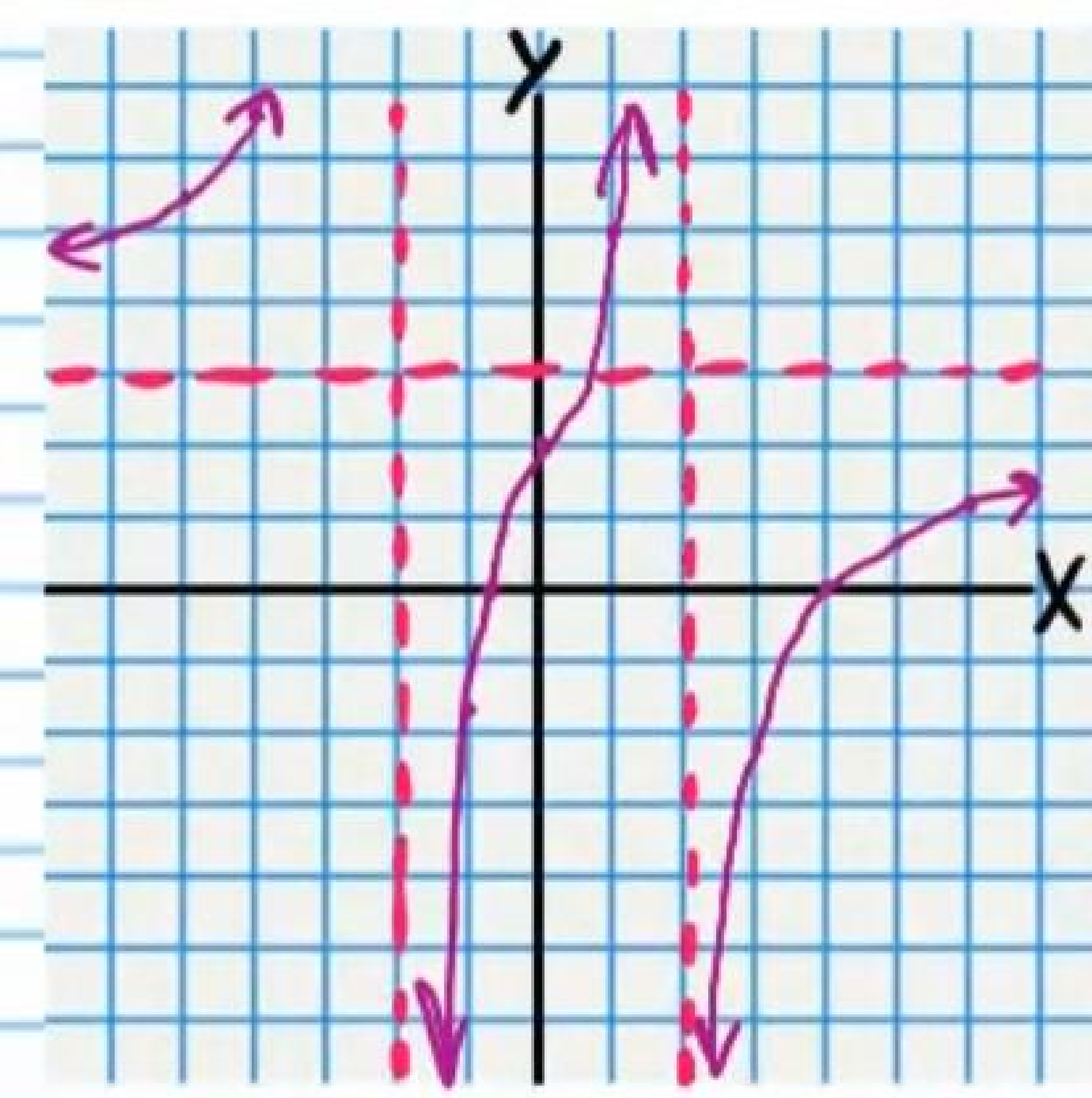
$\boxed{(0, 2)}$

X-Intercept

$\boxed{(4, 0)\ \&\ (-\tfrac{2}{3}, 0)}$

Sign Chart





③ $f(x)=\dfrac{x^4-1}{x^3-5x^2-x+5}$

Oblique Asymptote Equation

$\boxed{y=x+5}$

Hole(s)

$\boxed{x=-1,\ x=1}$

Vertical Asymptote Equations

$\boxed{x=5}$

Sign Chart

$-$ | $+$

5

Y-Intercept

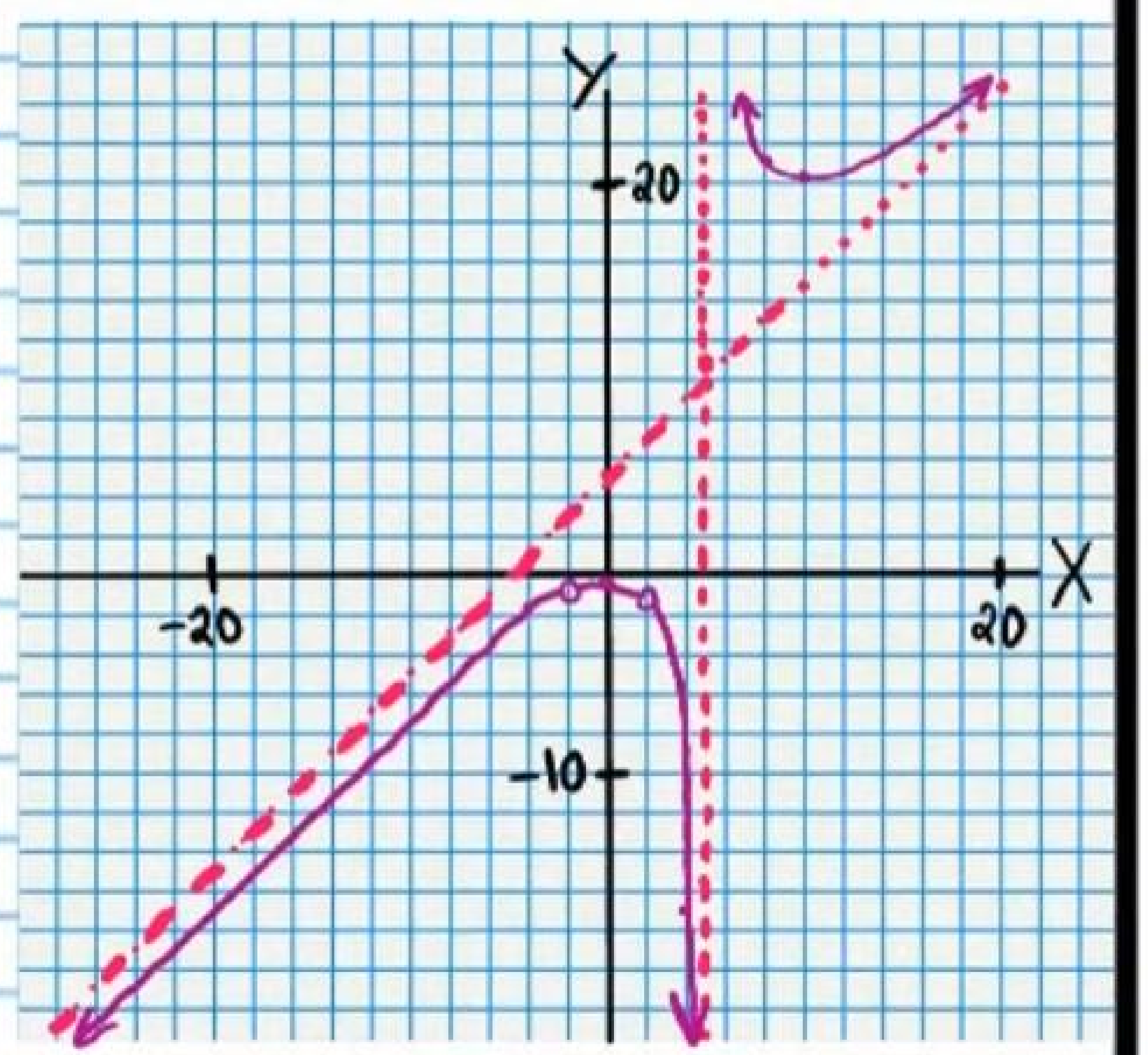
$\boxed{(0,-\tfrac{1}{5})}$

X-Intercept

$x^2+1=0$

$x=\pm i$

$\boxed{\text{None}}$



④ $f(x)=\dfrac{-x^2+3x-2}{x+3}$

Vertical Asymptote Equations

$\boxed{x=-3}$

Oblique Asymptote Equation

$\boxed{y=-x+6}$

Sign Chart

$+$ | $-$ | $+$ | $-$

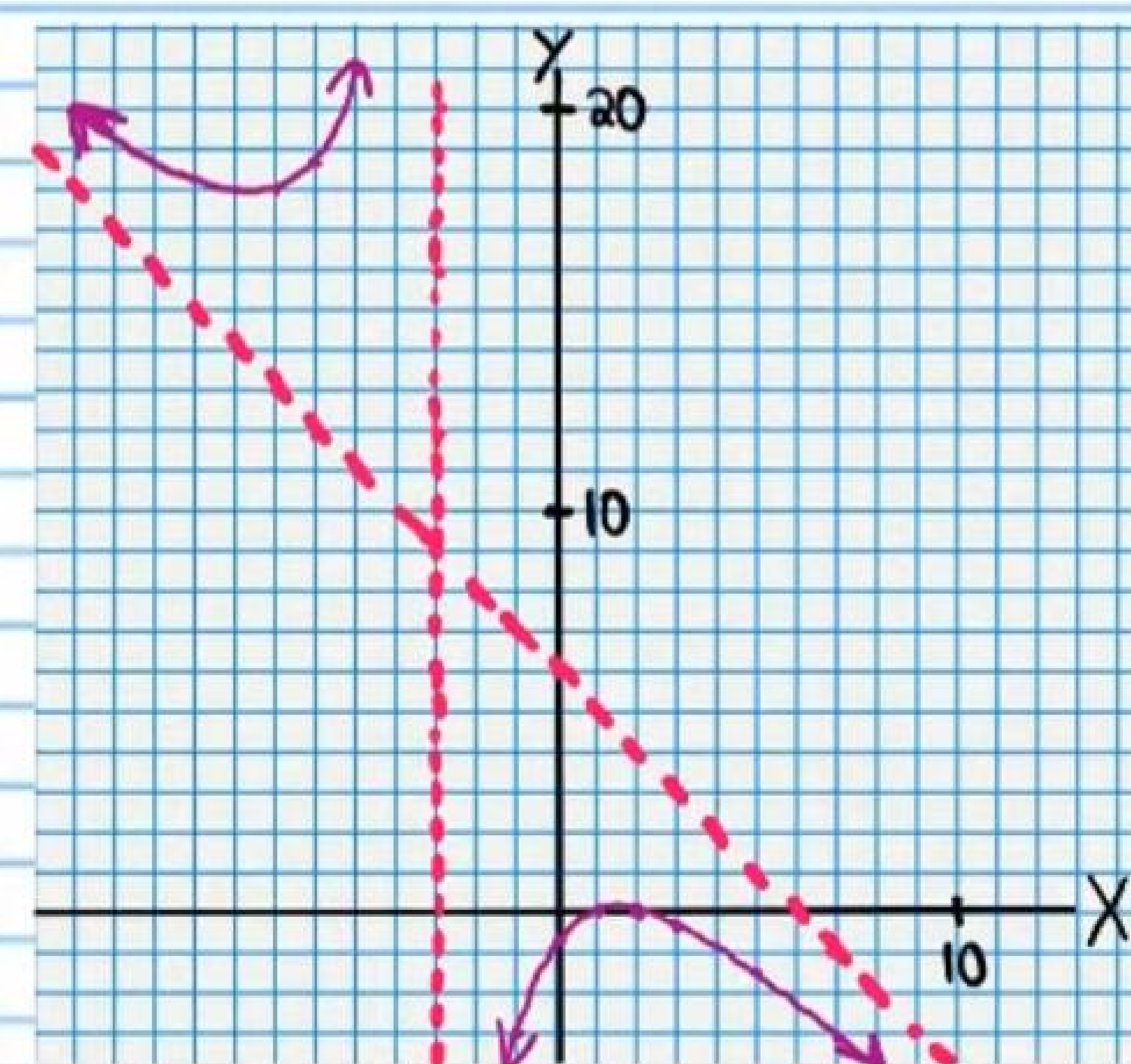
-3 1 2

Y-Intercept

$\boxed{(0,-\tfrac{2}{3})}$

X-Intercept

$\boxed{(2,0)\ \&\ (1,0)}$



⑤ $f(x) = \dfrac{x^6+2x^5}{x^4+5x^3+6x^2}$

Hole(s)

$\boxed{x=-2,\ x=0}$

$$\frac{x^5(x+2)}{x^2(x+2)(x+3)}$$

Vertical Asymptote Equation(s)

$\boxed{x=-3}$

Asymptote Curve

$\boxed{y=x^2-3x+9}$

Y-Intercept

$\boxed{\text{None}}$

X-Intercept

$\boxed{\text{None}}$

⑥ $f(x) = \dfrac{x^3+3x^2+x+3}{x-3}$

Y-Intercept

$\boxed{(0,-1)}$

X-Intercept

$\boxed{(-3,0)}$

Vertical Asymptote Equations

$\boxed{x=3}$

Asymptote Curve

$\boxed{y=x^2+6x+19}$